

CORE1A - BUILD UNDERSTANDING

Learn the Idea

Core1A turns validated obligations into a teaching sequence. It is not forced into the Core1B open-ended/self-help pattern.

This sample demonstrates:

- See the object before symbols.
- Bridge picture -> words -> coordinates -> equation.
- Derive the shortcut condition.
- Contrast the tempting wrong shortcut.
- Move worked -> guided -> independent.

Sample topic

A point on an axis equidistant from two fixed points

Generated review fixture - not an official exam source

1. See and represent

P can move only along the x-axis, so write $P=(x,0)$.



WORDS -> SYMBOLS

"P lies on x-axis" becomes $P=(x,0)$.

GEOMETRY -> EQUATION

$PA=PB$ becomes $(x-a)^2+b^2=(x-c)^2+d^2$.

MEANING

The equation describes the same geometric constraint in an algebraic representation.

2. Derive, contrast, practise

The learner sees why the shortcut works before using it independently.

DERIVATION

If $A=(a,b)$, $B=(c,b)$: vertical terms cancel, giving $x=(a+c)/2$.

CONTRAST

$A=(-1,4)$, $B=(7,2)$: $P=(3,0)$ fails because $PA^2=32$ and $PB^2=20$.

GENERAL METHOD

$(x+1)^2+16=(x-7)^2+4 \rightarrow x=9/4$.

GUIDED

$A=(0,5)$, $B=(6,1)$, P on x -axis. Start $P=(x,0)$. Check $P=(1,0)$.

INDEPENDENT

$A=(-4,3)$, $B=(2,1)$, P on y -axis. Find P . Check $P=(0,5)$.